Sensor Averaging

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April 2026

1 Introduction

Averaging multiple sensor readings of zero bias sensors reduces error, for example in GPS, if we connect to multiple GPS servers, and average the locations given, we can reduce error in position prediction.

X ground truth position of device

$\hat{X}_i$ position predicted from satellite i

$E[\hat{X}_i - X] = 0$ (0 bias for simple example)

$E[(\hat{X}_i - X)^2] = error_i$ $(variance_i)$

theorem 1: $E[(\sum \hat{X}_i)/n - X] = 0$

theorem 2: $E[((\sum \hat{X}_i)/n - X)^2] < error_i$

2 Theorem 1

$E[(\sum \hat{X}_i)/n - X] =$

$E[(\sum^n \hat{X}_i)/n - nX/n] =$

$E[(\sum^n \hat{X}_i)/n - (\sum^n X)/n] =$

$E[1/n(\sum^n \hat{X}_i - \sum^n X)] =$

$E[1/n \sum^n (\hat{X}_i - X)] =$

$1/\text{n}\sum^n E[\hat{X}_i - X] = 1/n \cdot n \cdot 0 = 0$

3 Theorem 2

$E[((\sum \hat{X}_i)/n - X)^2] < error_i$

$E[((\sum \hat{X}_i)/n - X)^2] =$

$E[((\sum \hat{X}_i)/n - nX/n)^2] =$

$E[((\sum \hat{X}_i)/n - (\sum X)/n)^2] =$

$1/n^2\ E[((\sum \hat{X}_i) - (\sum X))^2] =$

$1/n^2\ E[(\sum(\hat{X}_i - X))^2] =$

$1/n^2\ E[(\ (\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \ldots + (\hat{X}_n - X)\ )^2] =$

$$
\begin{aligned}
1/n^2\ E[\quad & (\hat{X}_1 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \ldots + (\hat{X}_n - X))+ \\
& (\hat{X}_2 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \ldots + (\hat{X}_n - X))+ \\
& (\hat{X}_3 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \ldots + (\hat{X}_n - X))+ \\
& \qquad\qquad\qquad \vdots \\
& (\hat{X}_n - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \ldots + (\hat{X}_n - X))\ \ ] =
\end{aligned}
$$

$1/n^2\ E[\ ((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X))\ n\ ] =$

$1/n\ E[\ ((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X))\ ] =$

Now we need the following lemma:
X_i and X_j independent gives $E[(\hat{X}_i - X)(\hat{X}_j - X)] = 0$

proof:

$E[(\hat{X}_i - X)(\hat{X}_j - X)] =$

$E[(\hat{X}_i\hat{X}_j - \hat{X}_iX - X\hat{X}_j - X^2] =$

$$E[\hat{X}_i\hat{X}_j] \quad - \quad E[\hat{X}_i]X \quad - \quad XE[\hat{X}_j] \quad - \quad X^2 =$$

$(\hat{X}_i, \hat{X}_j$ independent)

$$E[\hat{X}_i]E[\hat{X}_j] \quad - \quad E[\hat{X}_i]X \quad - \quad XE[\hat{X}_j] \quad - \quad X^2 =$$

$$X \cdot X \quad - \quad X \cdot X \quad - \quad X \cdot X \quad - \quad X^2 = 0$$

($E[\hat{X}_i] = X$ for simplicity we assume zero bias as stated above.)

Now, continuing with the proof using the lemma $E[(\hat{X}_i - X)(\hat{X}_j - X)] = 0$

$1/n\ E[\ ((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X))\ ] =$

$1/n\ E[\ ((\hat{X}_i - X)^2] < error_i = E[(\hat{X}_i - X)^2]$

If we normalize the error (like a standard deviation), we see

$(1/n\ E[\ ((\hat{X}_i - X)^2])^{1/2} = (1/n)^{1/2}\ (E[\ ((\hat{X}_i - X)^2])^{1/2}$,

This is nice, as it shows the expected normalized error of n sensors averaged is $\sqrt{1/n}$ times the single sensor normalized error. If we want around half the error we can use 4 channels, if we want around one fourth the error, we can use 16 channels, and so on.